

Physiology: Cardiac Action Potentials

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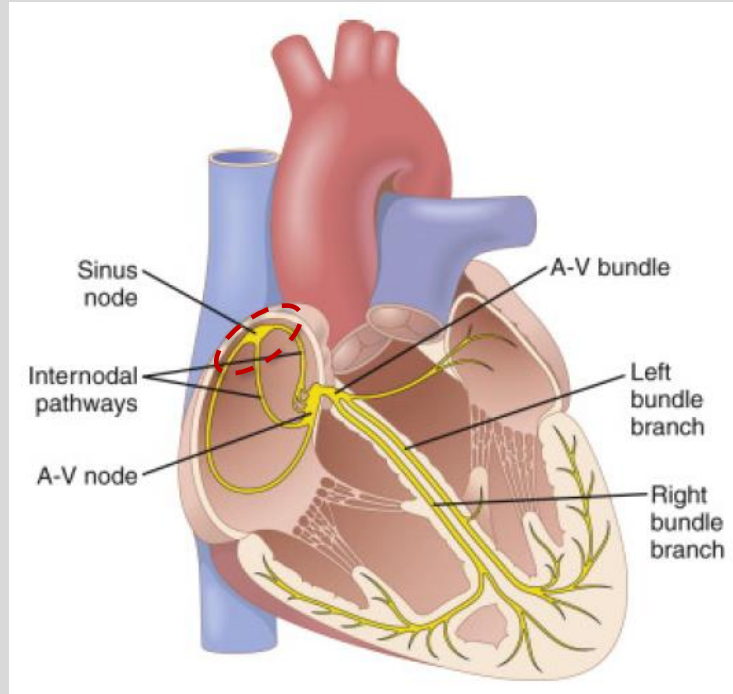
Session Objectives

- Describe the two types of cardiac action potentials and their correlation to specific types of cardiac myocytes
- Explain different phases and their ionic currents underlying the two types of cardiac action potentials
- Analyze the processes of cardiac excitation-contraction coupling
- Interpret the effective refractory period of cardiac myocytes, its determinants and its physiological importance
- Explain how changes in ion currents affect cardiac action potentials, and predict the effects of hypokalemia and hyperkalemia on action potentials

Heart & Specialized Excitatory and Conduction System

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❖ Normal heartbeat is initiated in the SA node and transmitted throughout the heart in the form of cardiac action potential (AP)

❖ AP is the basis for all cardiac electrophysiology

-ECG

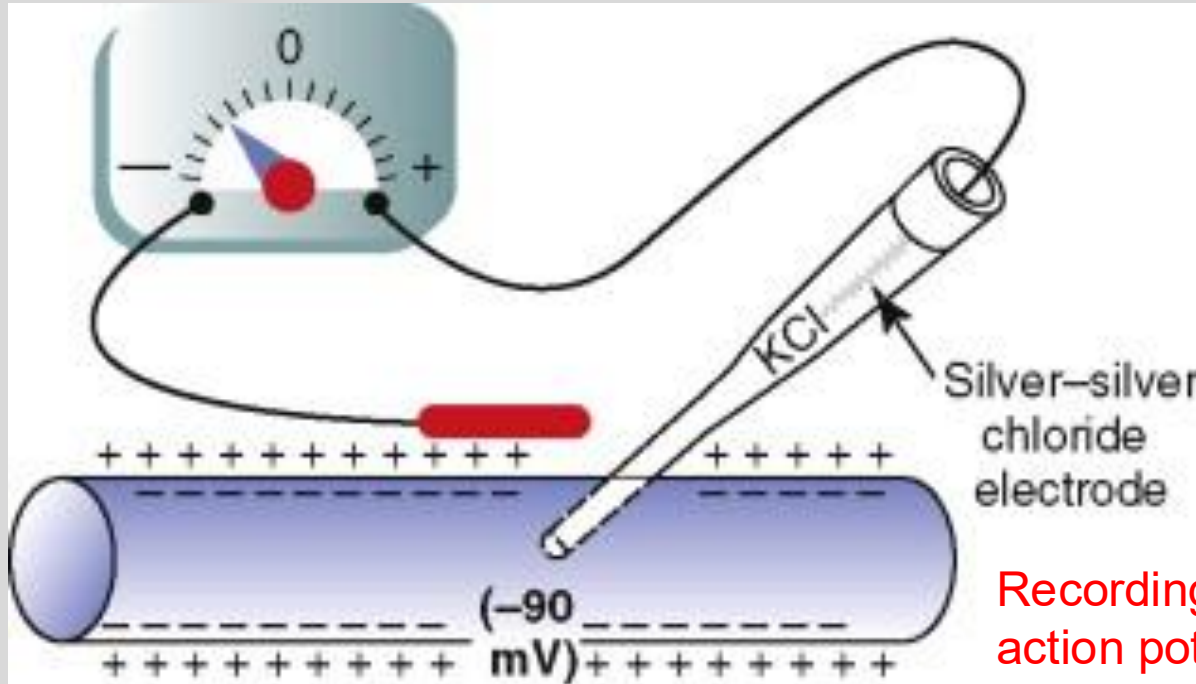
-Arrhythmias & medications

-Clinical EP

Action Potential Recording

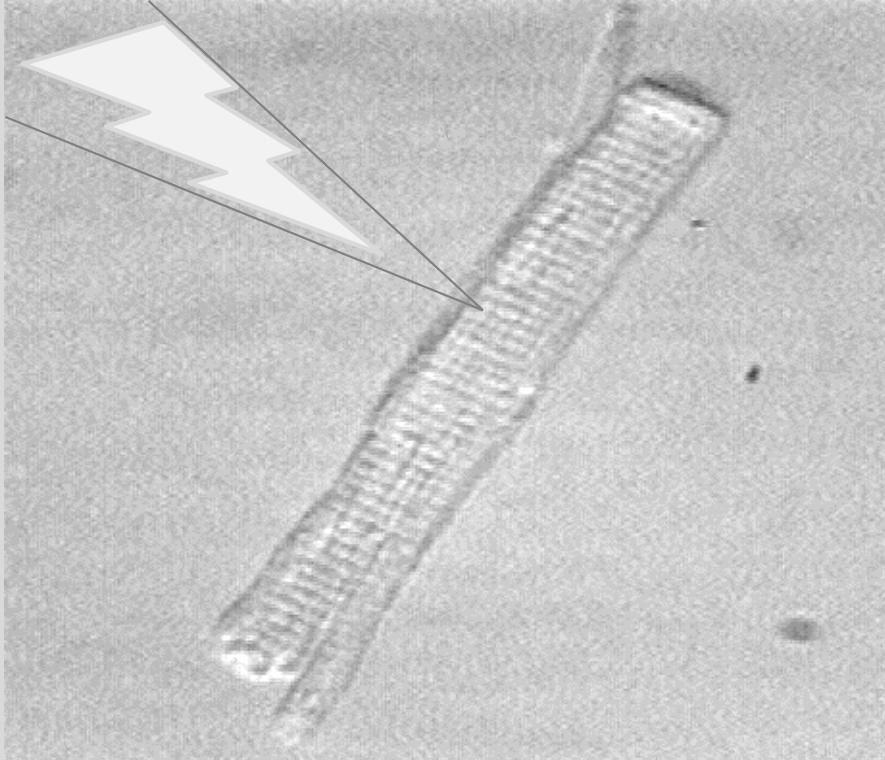
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Recording of intracellular
action potential (AP)

An isolated myocyte



Our own data

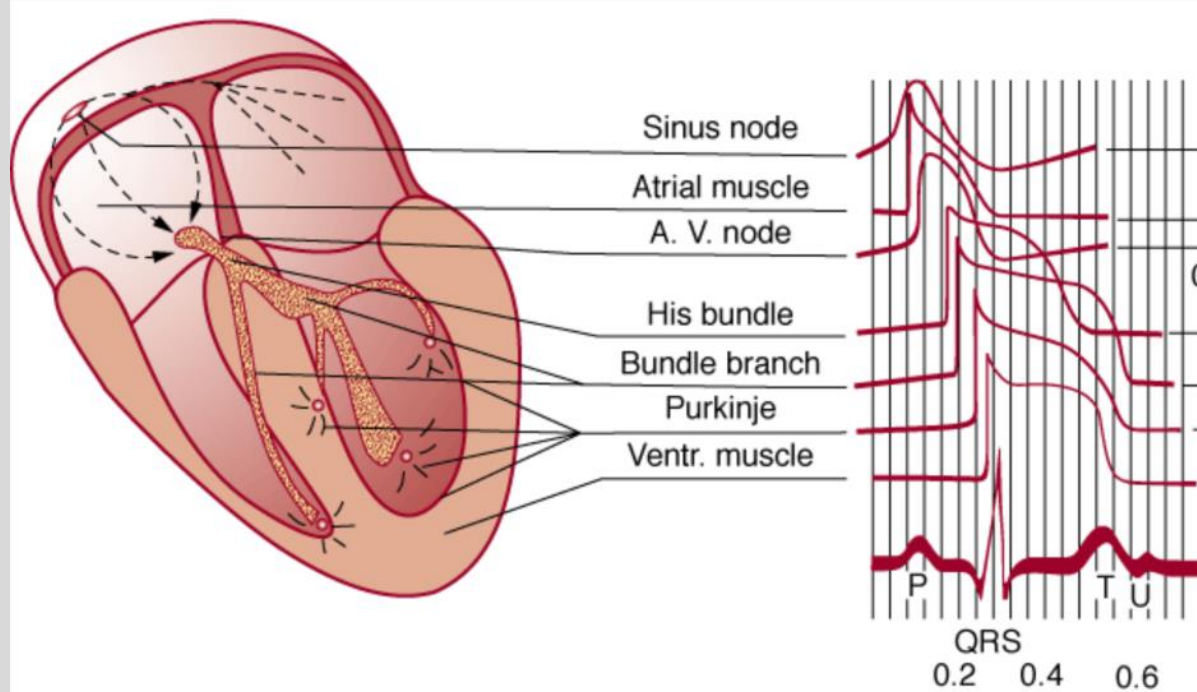
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Action potentials from different cardiac fibers

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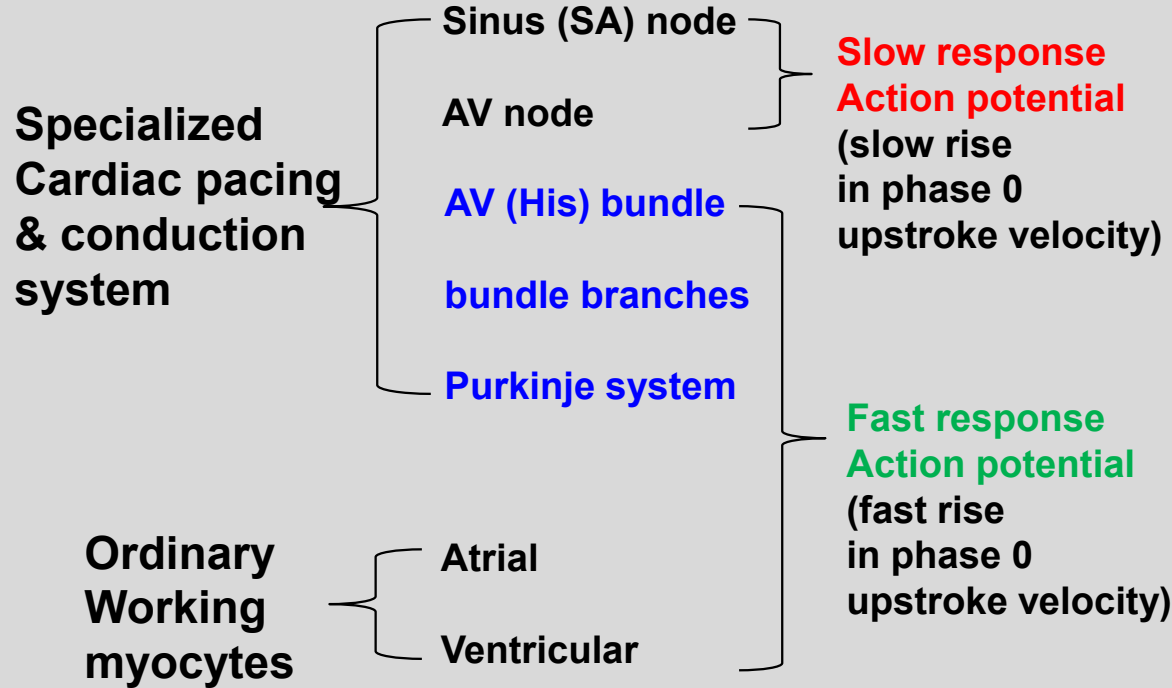


Source: Fuster V, Walsh RA, Harrington RA: *Hurst's The Heart*,
13th Edition: www.accessmedicine.com
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Cardiac myocytes and 2 types of cardiac action potentials

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Fast versus Slow APs

	Fast response AP	Slow response AP
Cause of Phase 0 Depolarization	Opening of Na ⁺ Channel	Opening of Ca ²⁺ (L) Channel
Speed of depolarization (dv/dt)	Fast (>100 v/s)	Slow (~10 v/s)

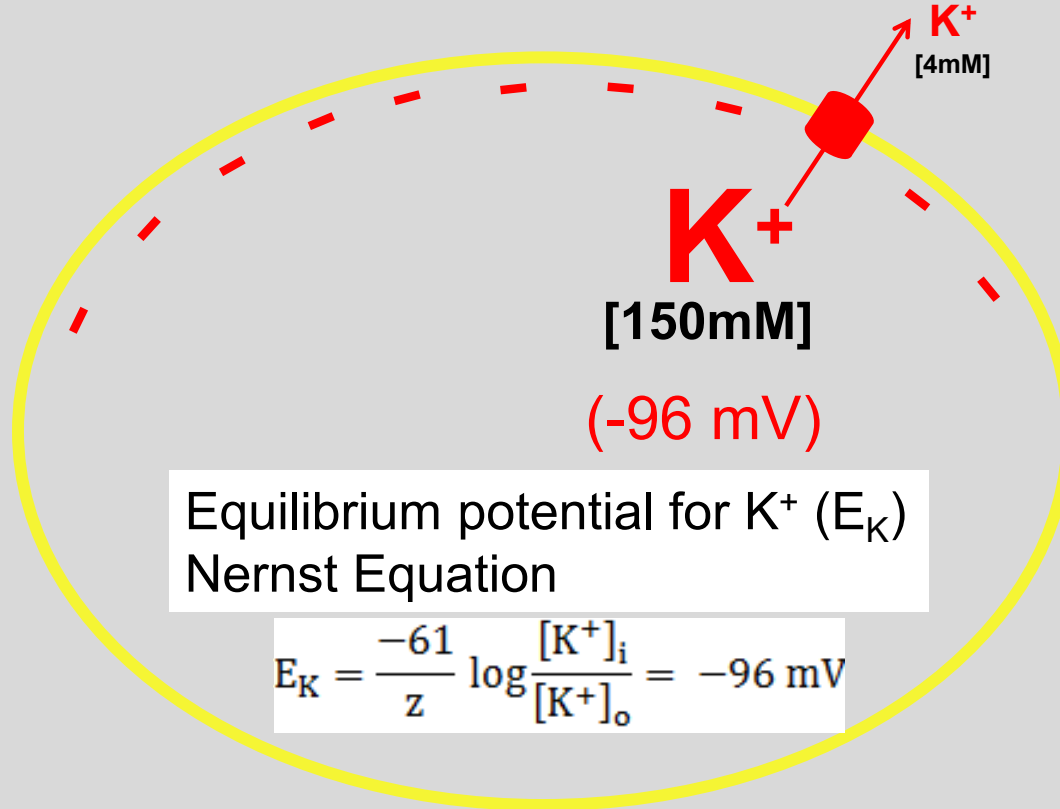
Major Elements for Cardiac AP

1. **3 major ions: K^+ , Na^+ and Ca^{++}**
concentration gradient across the cell membrane
(others, like Cl^- , play a minor role)
2. **Relative membrane permeability** to
 K^+ , Na^+ and Ca^{++}
(regulated by highly selective **ion channels**)
3. **Electrogenic ion pumps**
 Na^+/K^+ ATPase
 Na^+/Ca^{++} exchanger
 Ca^{++} -ATPase (SERCA, sarco-endoplasmic reticulum calcium-ATPase)

Potassium Diffusion Potential

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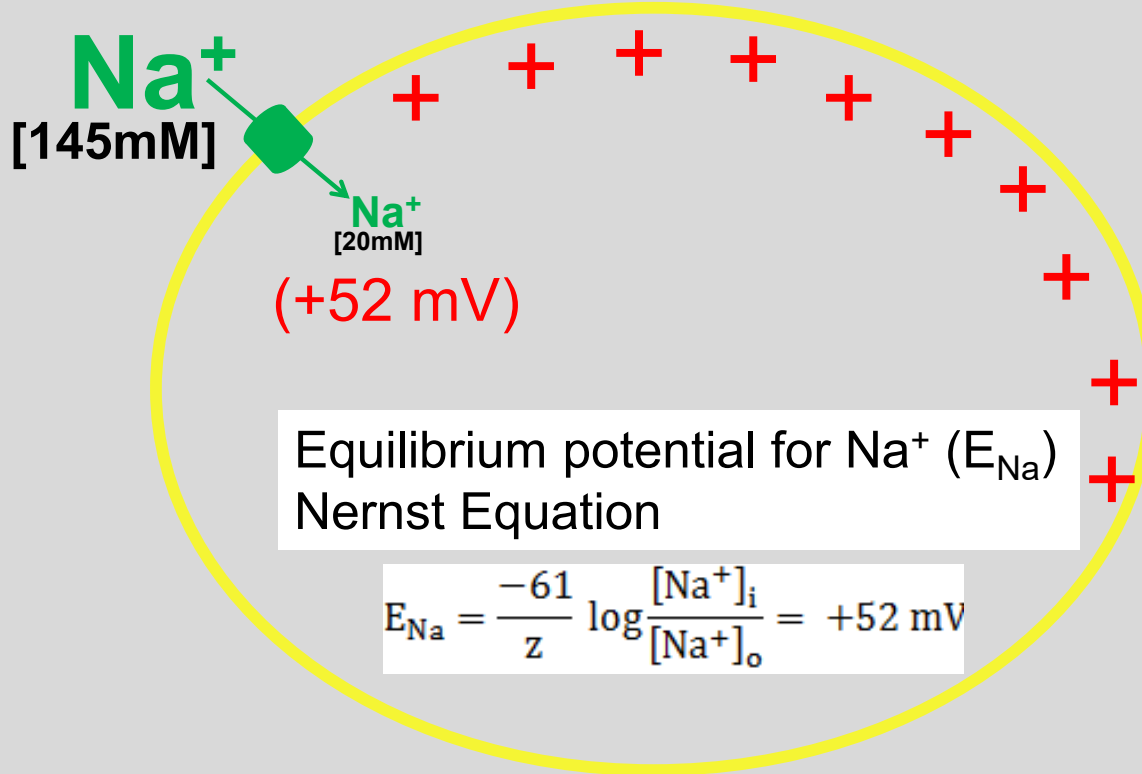
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Sodium Diffusion Potential

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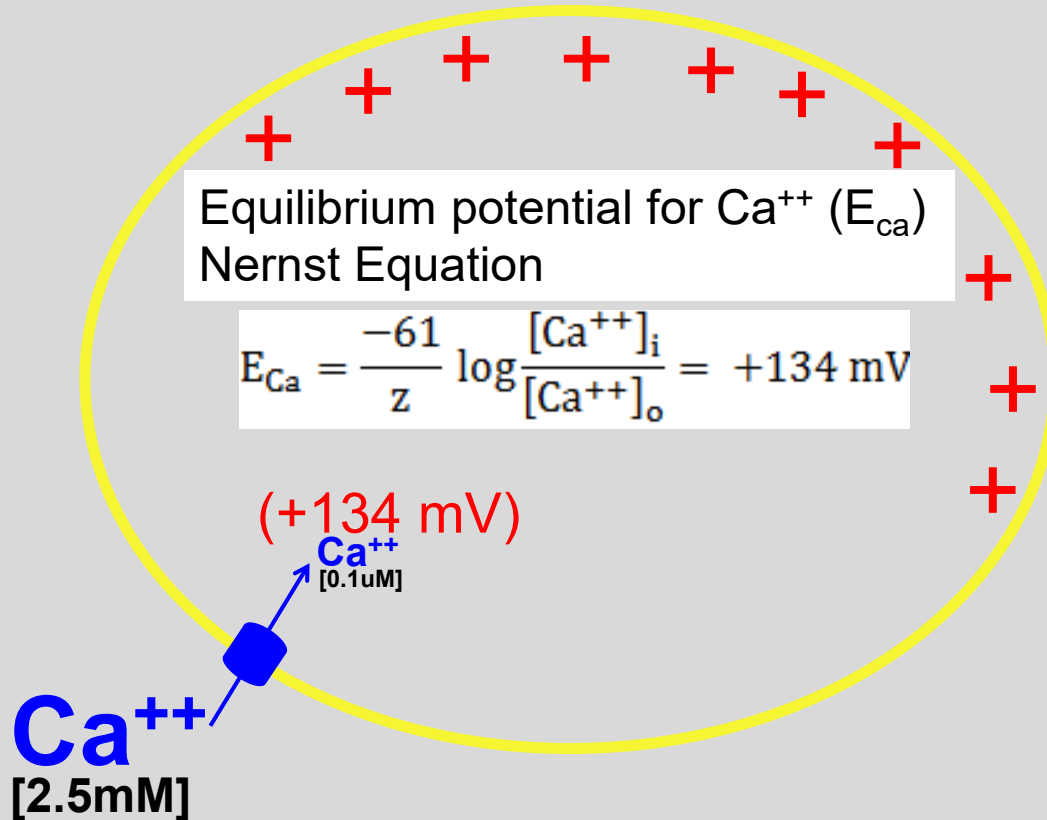
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Calcium Diffusion Potential

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Membrane Potential Equation

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$$E_m = \frac{gK(E_K) + gNa(E_{Na}) + gCa(E_{Ca}) + \dots}{gK + gNa + gCa + \dots}$$

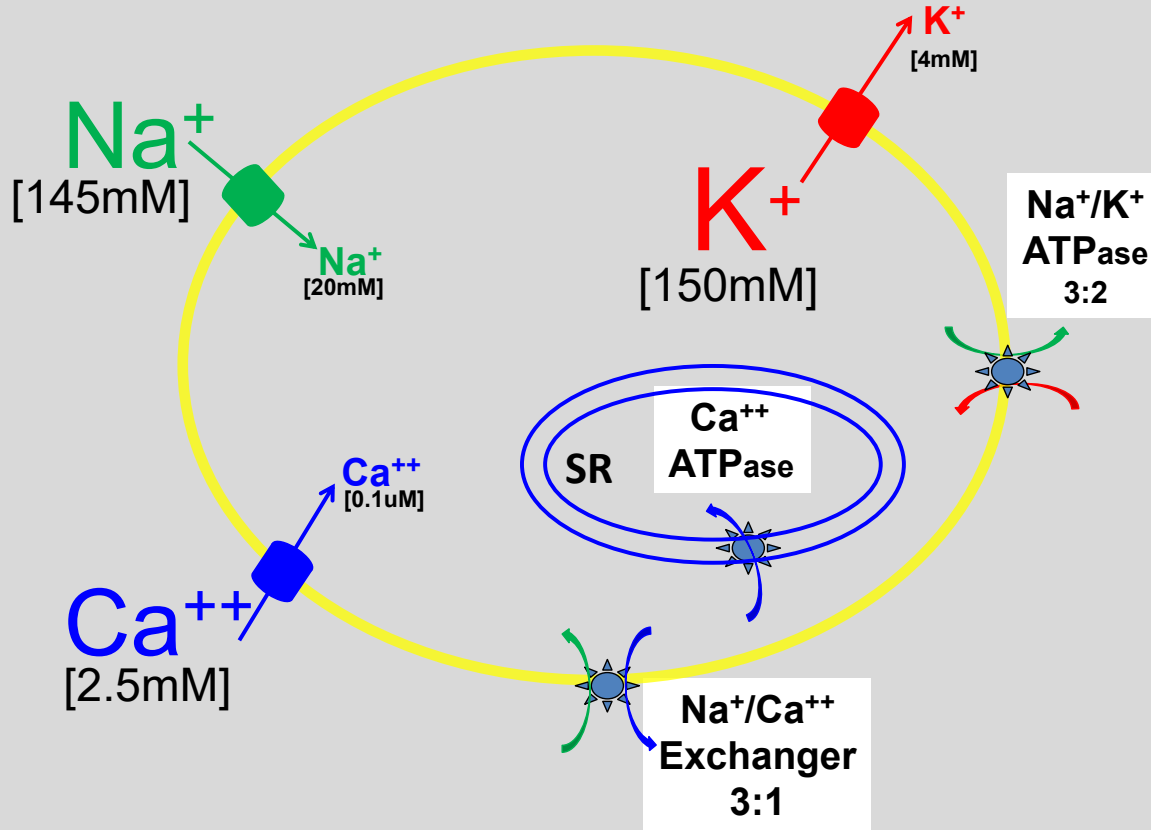
$$E_m = g'K(E_K) + g'Na(E_{Na}) + g'Ca(E_{Ca}) + \dots$$

$$E_m = g'K(-96) + g'Na(+52) + g'Ca(+134) + \dots$$

Electrolytes Homeostasis

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Fast Response Action Potential

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5 phases

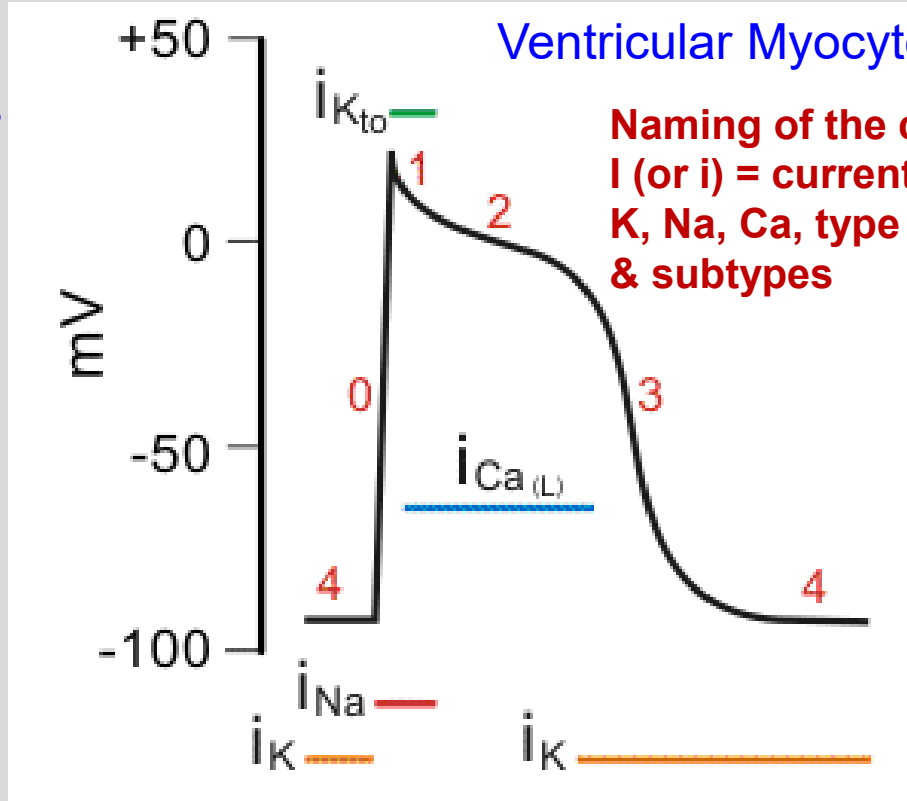
Phase 0

Phase 1

Phase 2

Phase 3

Phase 4

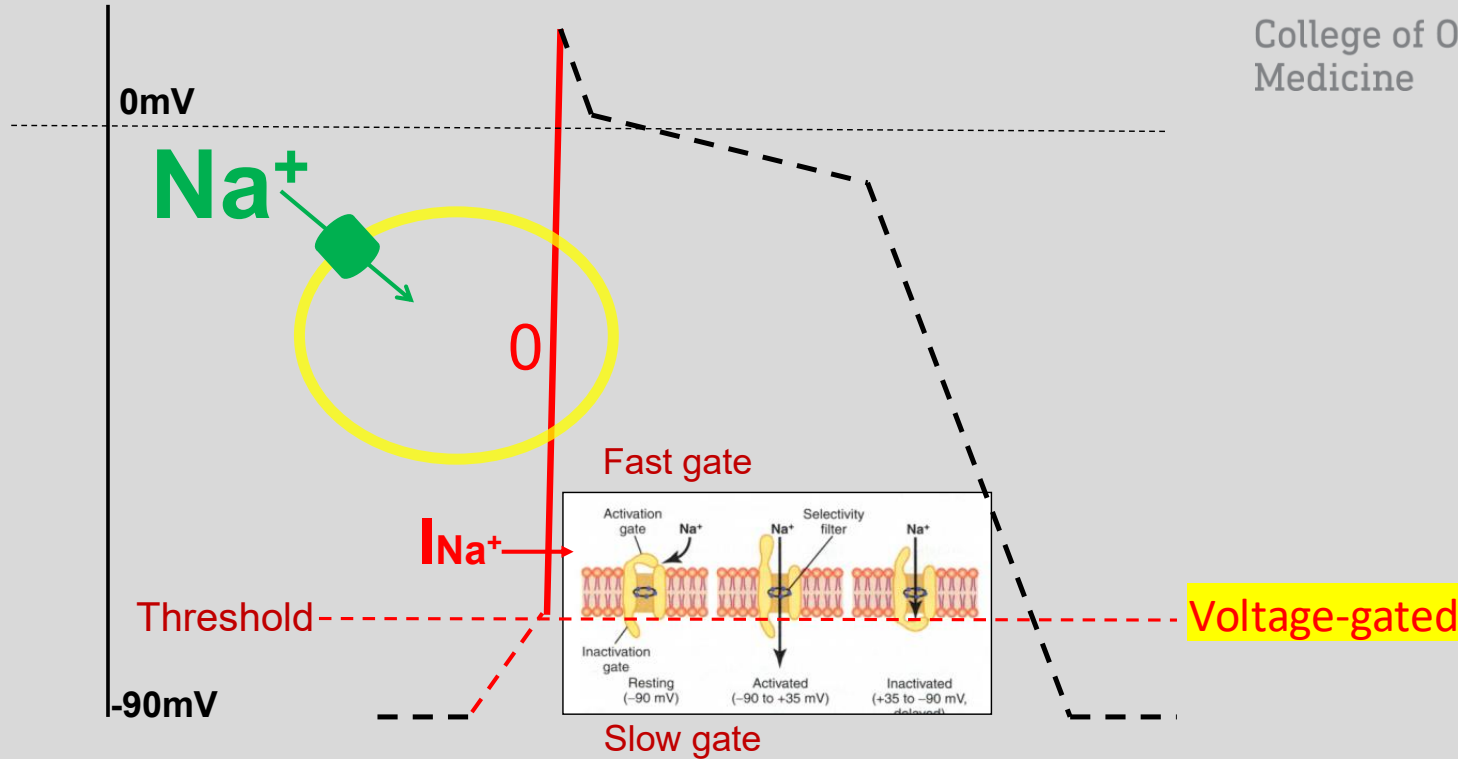


Depolarization (fast rise, high dv/dt)

Phase 0

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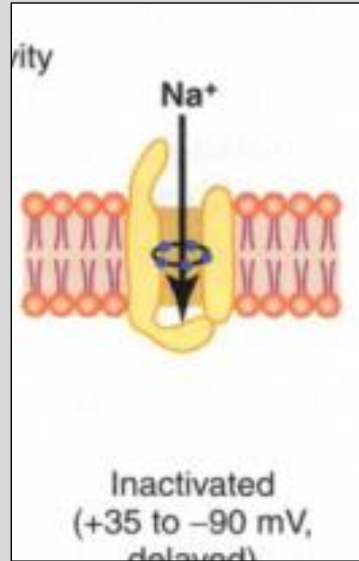
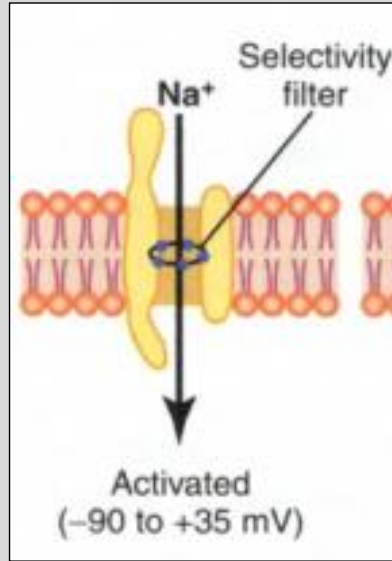
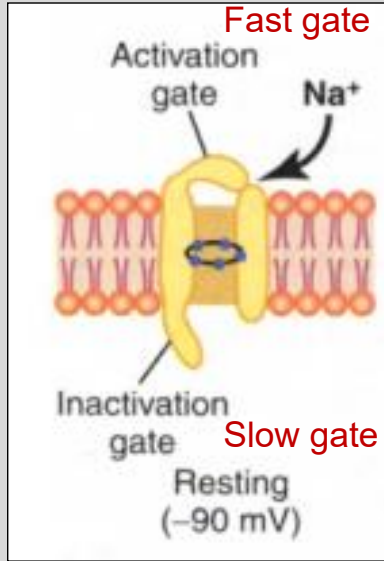
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Na⁺ channel status in different phases

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4

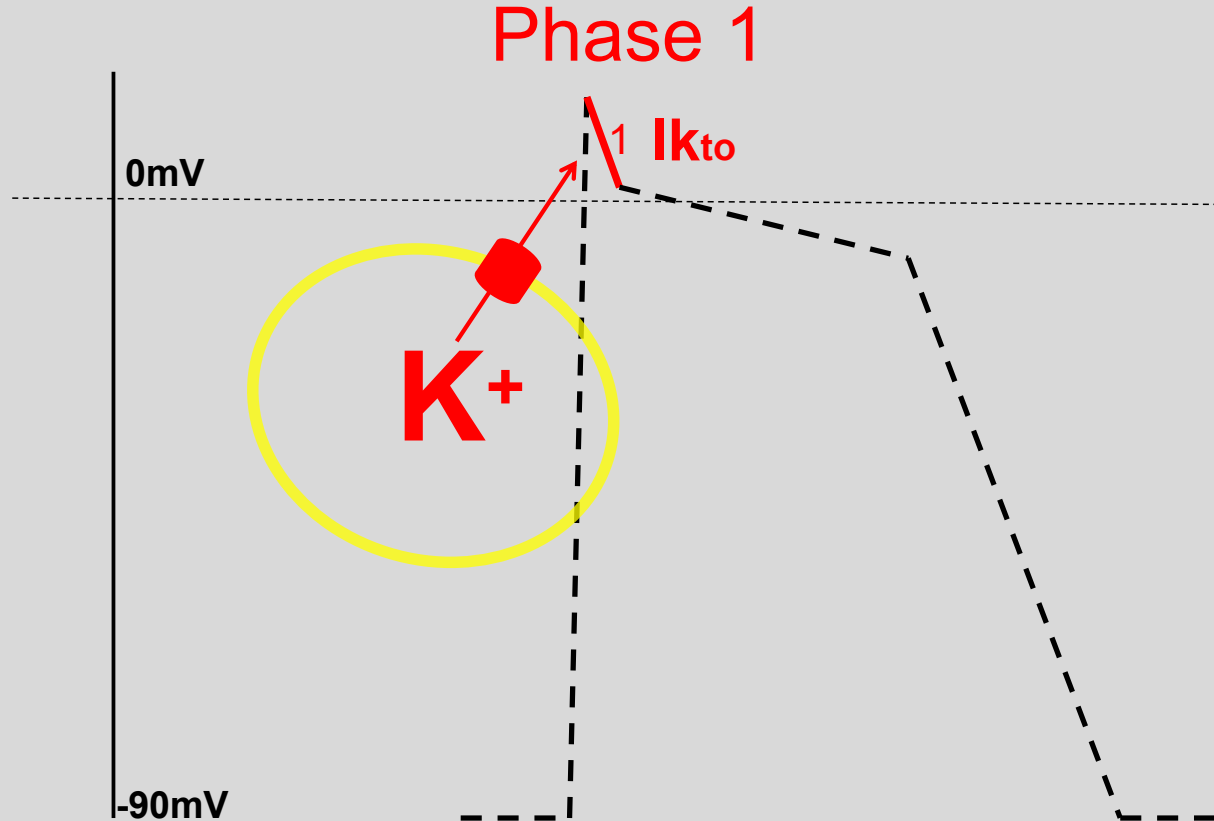
0

1-3

Early Repolarization

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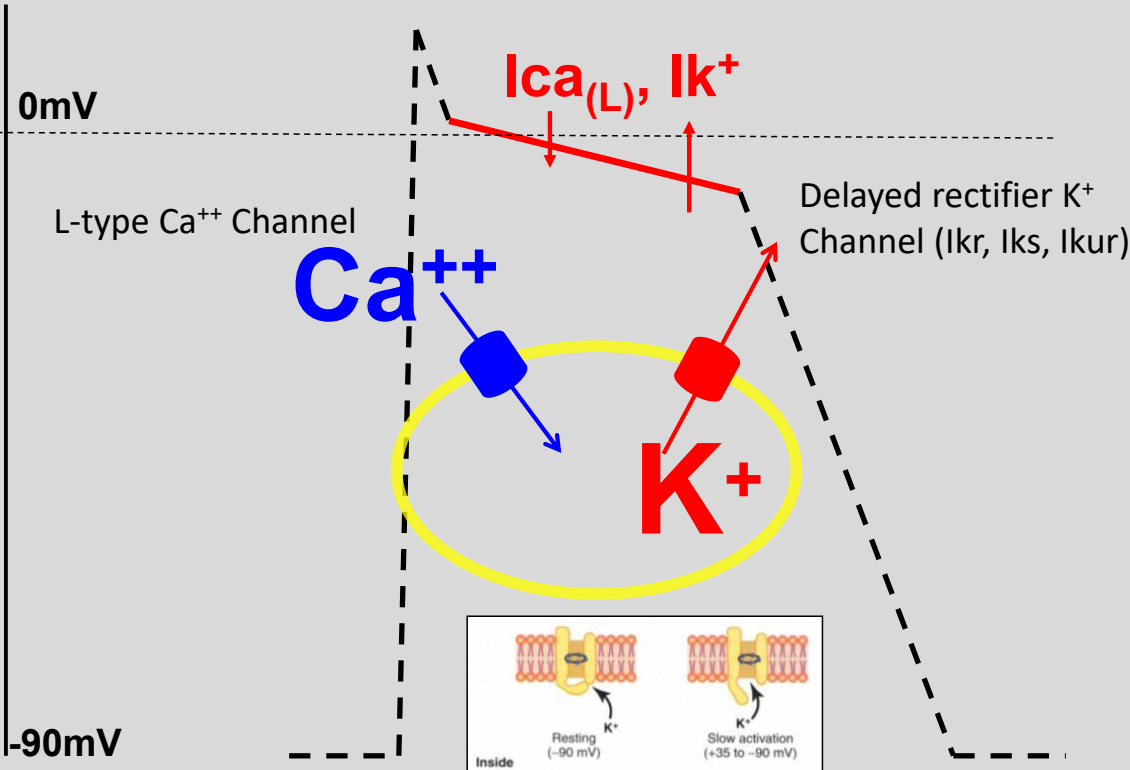


Plateau Phase

Phase 2

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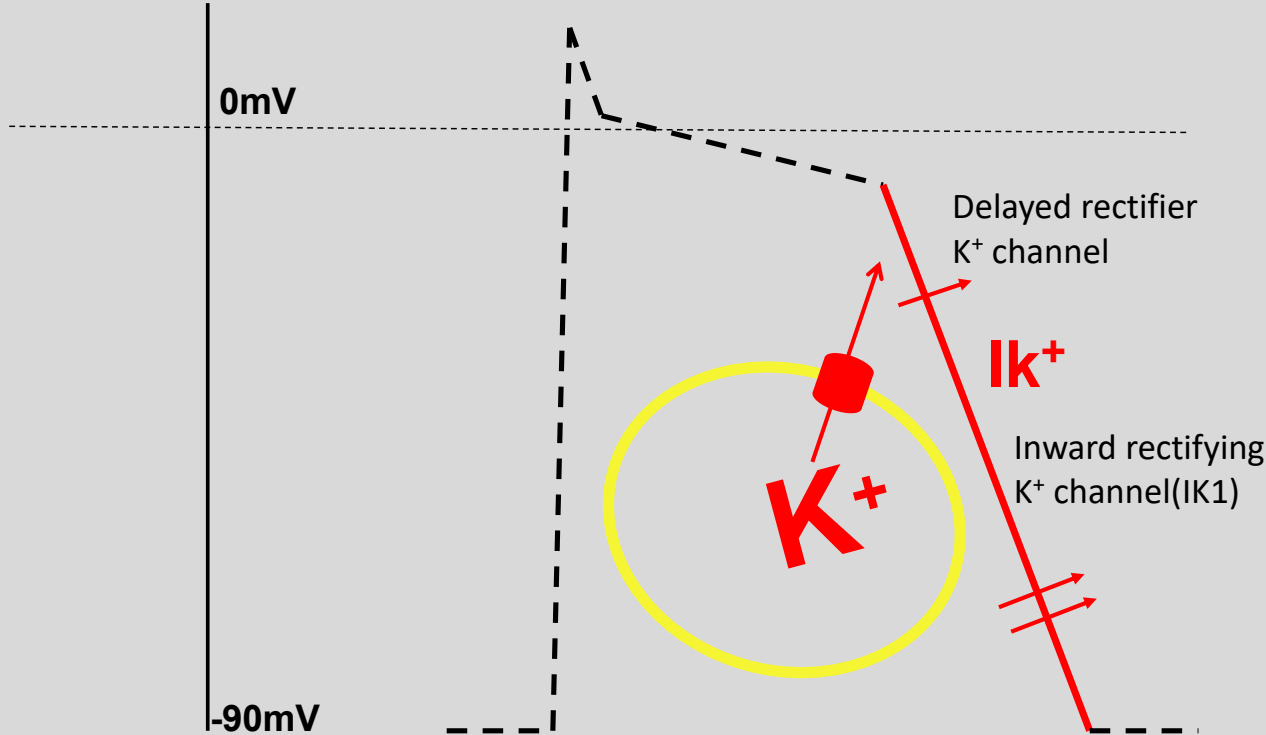
Potassium channel illustration from Guyton and Hall Textbook of Medical Physiology, e13, Figure 5-7

Rapid repolarization

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Phase 3



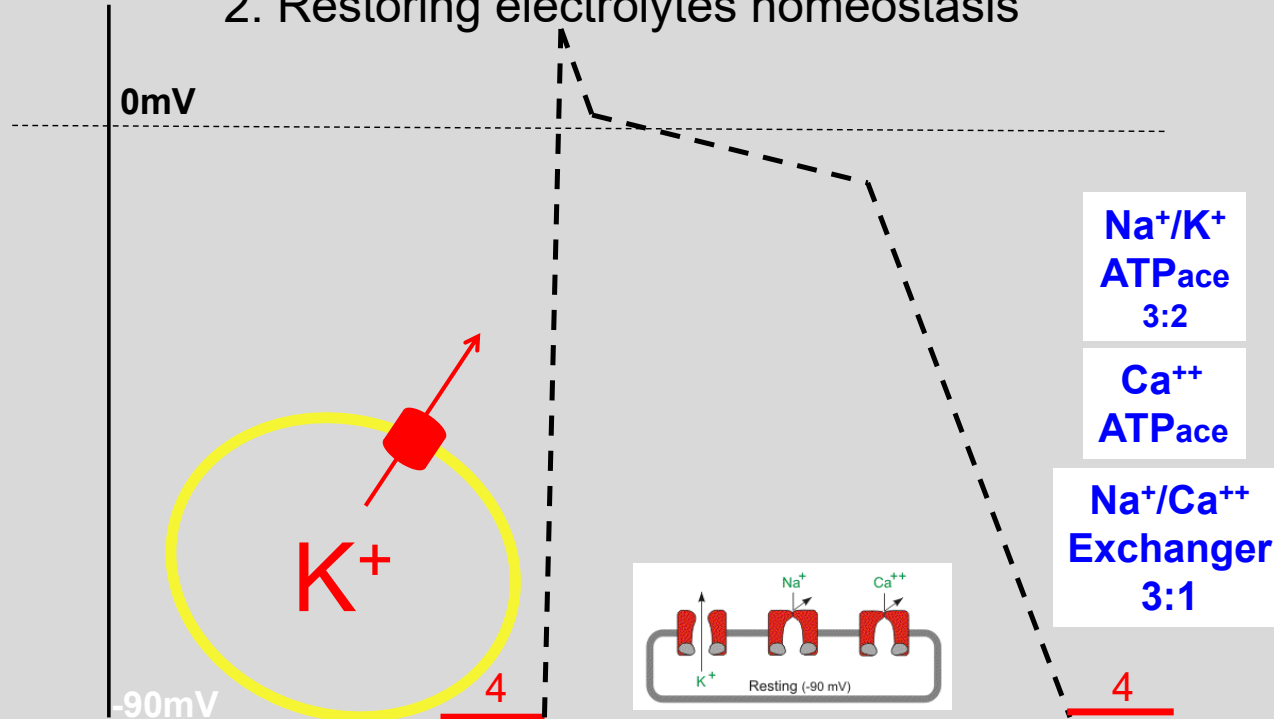
Resting Membrane Potential

(Phase 4)

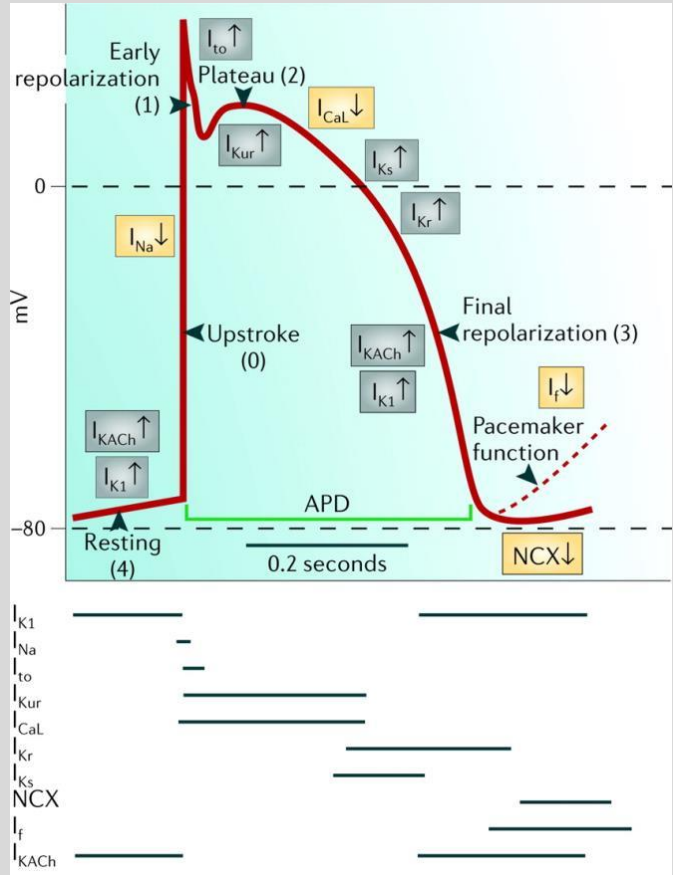
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1. Maintaining resting membrane potential
2. Restoring electrolytes homeostasis



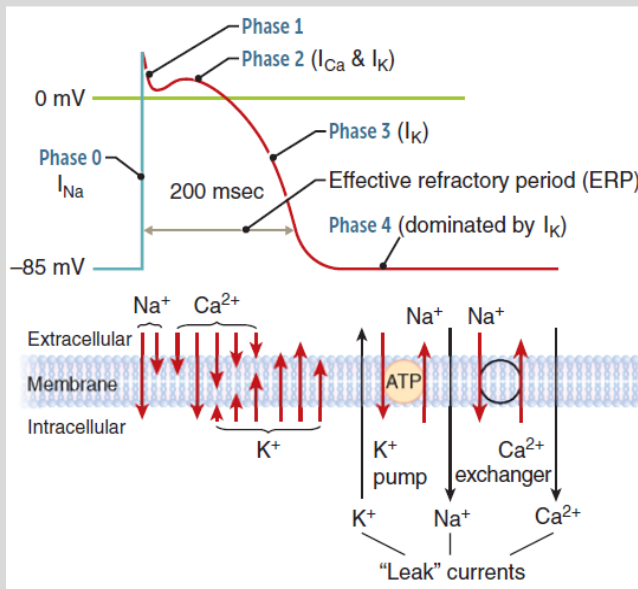
Membrane currents & action potential



Minimum requirement

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Phase 0 = rapid upstroke (depolarization)

—voltage-gated Na^+ channels open.

Phase 1 = initial repolarization

—inactivation of voltage-gated Na^+ channels

—opening of voltage-gated K^+ channels

Phase 2 = plateau

— Ca^{2+} influx through voltage-gated Ca^{2+} channels balances K^+ efflux.

Phase 3 = rapid repolarization

—opening of voltage-gated slow K^+ channels and closure of Ca^{2+} channels

Phase 4 = resting potential

—high K^+ permeability through K^+ channels

K⁺ (Hyperkalemia) and Cardiac AP

Normal
(3.5-5mM)

$$E_k = -61\text{mV} \log \frac{[K^+]_{\text{in}}}{[K^+]_{\text{out}}}$$
$$= -61\text{mV} \log \frac{150\text{mM}}{4\text{mM}}$$
$$= -96\text{mV}$$

Hyperkalemia $E_k = -61\text{mV} \log \frac{150\text{mM}}{7\text{mM}}$
 $= -81\text{mV}$

1. Membrane potential becomes more positive:

- increases excitability
- ↑membrane potential slows Na channel recovery (slows phase 0 depolarization, widens QRS on ECG)
- if stays above threshold, Na channel can't recover → cardiac arrest

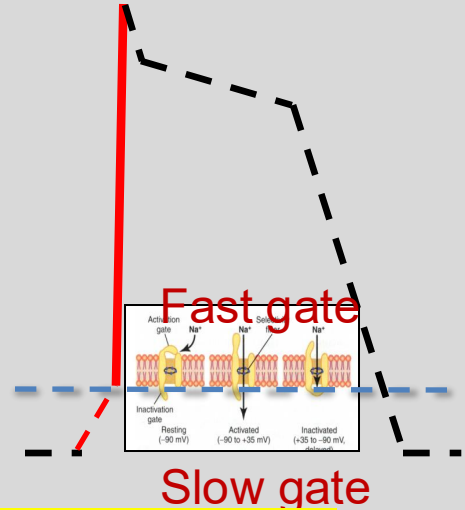
2. Increases K⁺ permeability (by open more K channels)

→ speeds up repolarization → shortens AP duration (ECG change:

Increases T wave amplitude or peaked T wave, widens QRS, VF or cardiac arrest)

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K⁺ (hyperkalemia) and cardiac arrest

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Calculated by Copilot (June 2026)

✓ Final Answer

A blood (extracellular) potassium concentration of roughly 17–18 mM would shift the cardiac myocyte resting membrane potential to about -55 mV. (the threshold level)

- ~17–18 mM represents **severe hyperkalemia**, typically incompatible with life
- At ~-55 mV, many cardiac Na⁺ channels become inactivated → **loss of excitability**, risk of **arrhythmias** or cardiac arrest

The number could slightly change depending on the assumption of intracellular K level

Clinical application: K⁺ concentration in cardioplegia solutions is around this level or slightly higher to arrest the heart during cardiac surgery or heart transplantation

K⁺ (Hypokalemia) and Cardiac AP

Hypokalemia

$$E_k = -61\text{mV} \log \frac{150\text{mM}}{1.5\text{mM}}$$
$$= -122\text{mV}$$

1. Moves potential to a more negative level
2. Decrease K⁺ permeability (fewer channels open)
Delays repolarization (increases AP duration, prolonged Q-T interval, increased risk of ventricular arrhythmia)
(ECG change: flattened T wave, appearance of u wave, Q-T prolongation)

K⁺ concentrations on AP

Canine ventricular AP and ECG

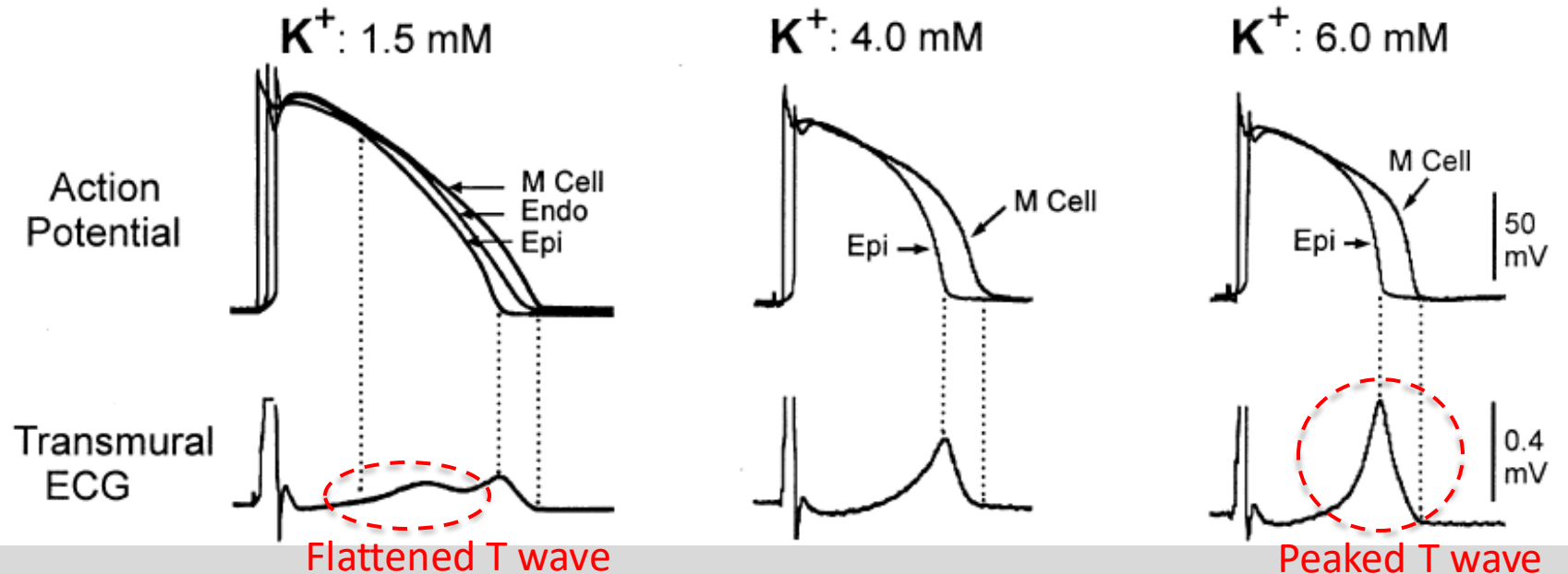
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Hypokalemia

Normal

Hyperkalemia

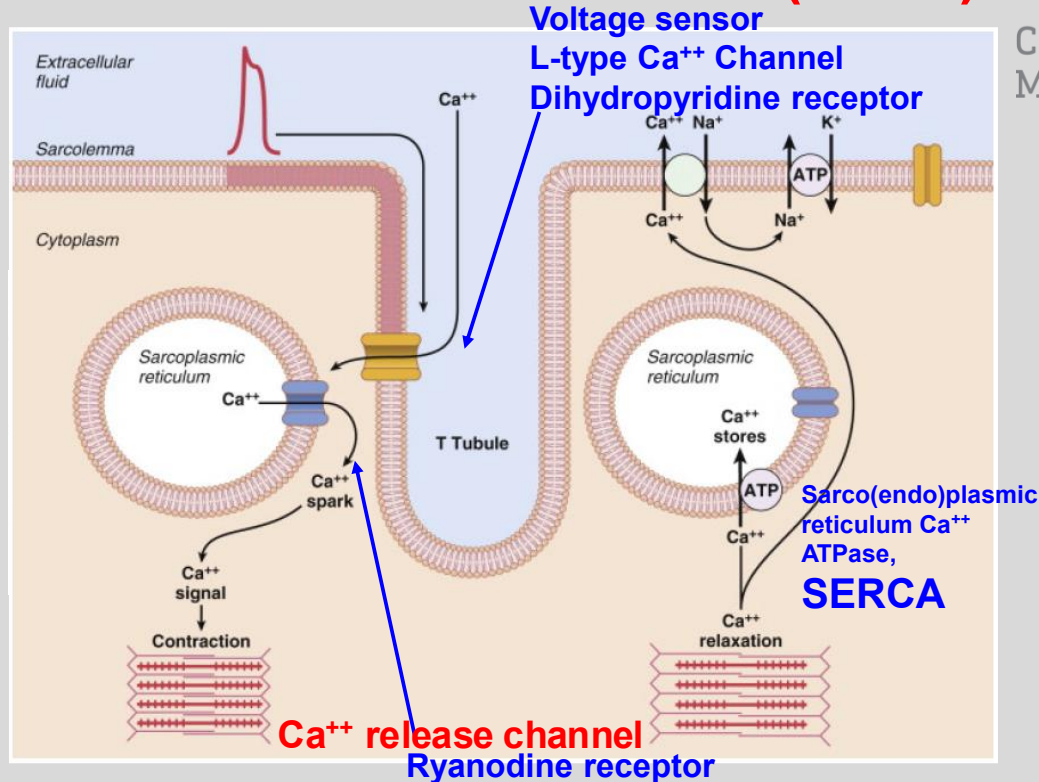


Excitation-Contraction Coupling

Calcium-induced Ca^{++} release (CICR)

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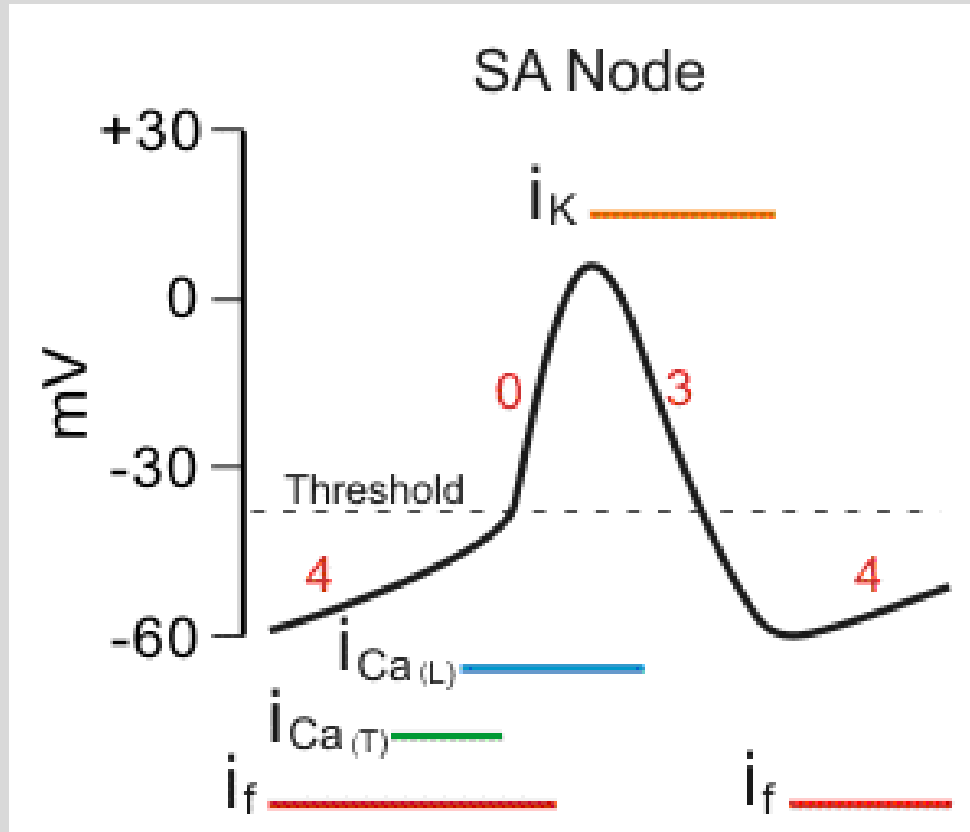
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Slow Response Action Potential

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(Pacemaker cell)

3 phases

Phase 0

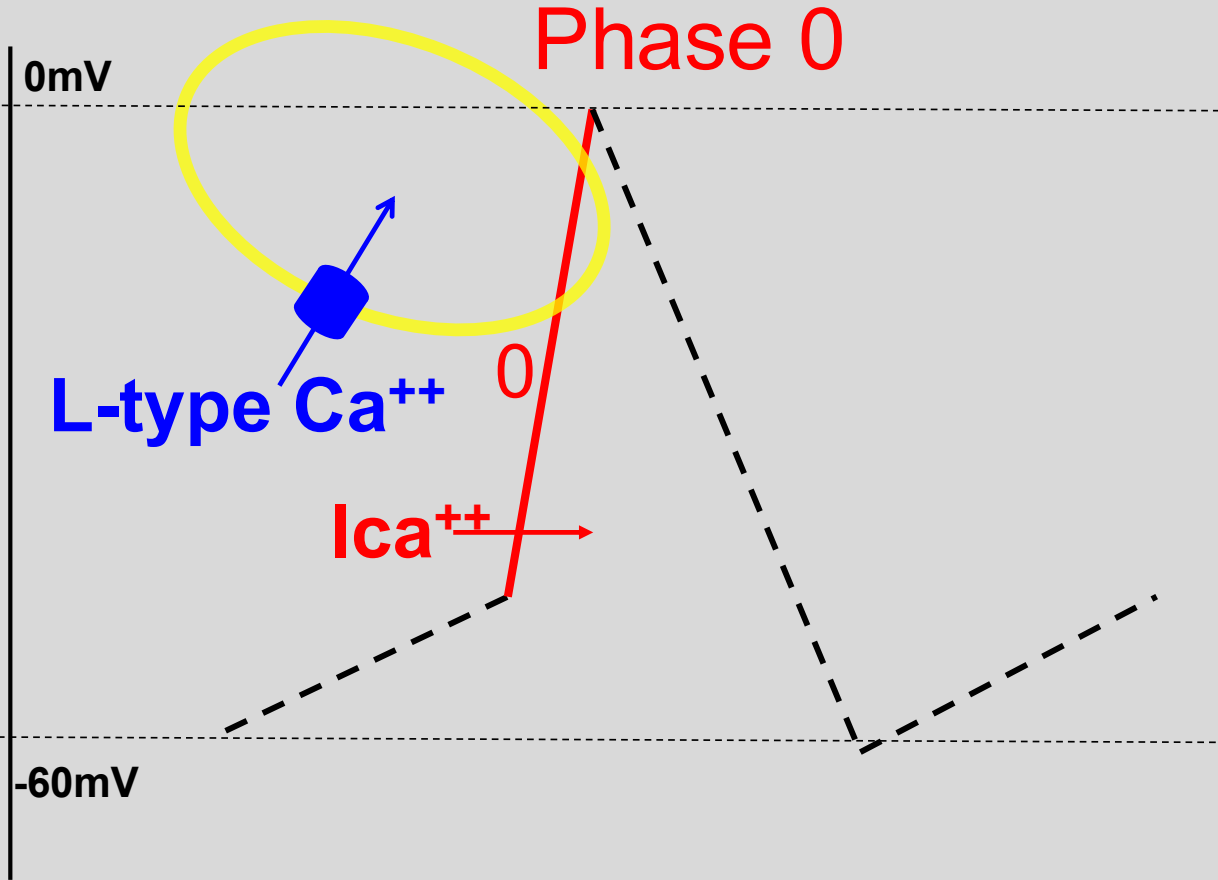
Phase 3

Phase 4

Depolarization (slow rise, low dv/dt)

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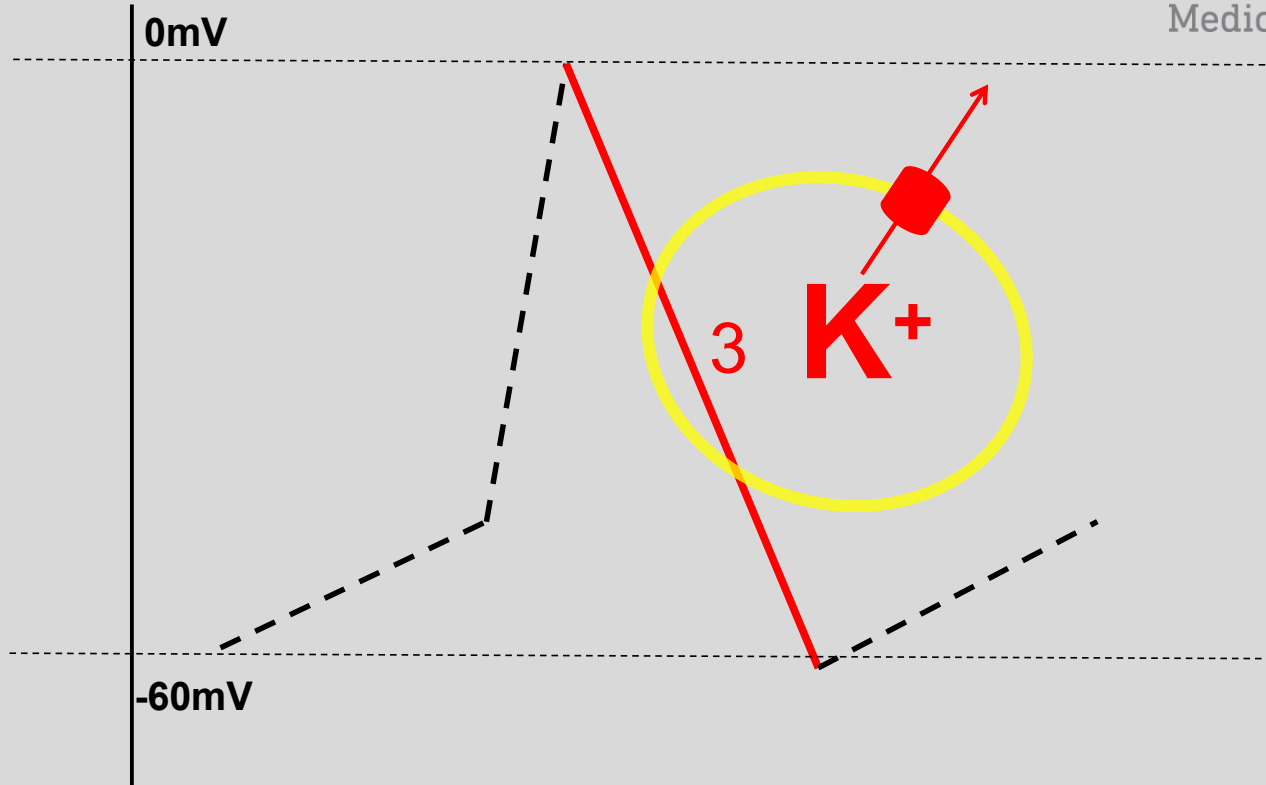


Repolarization

Phase 3

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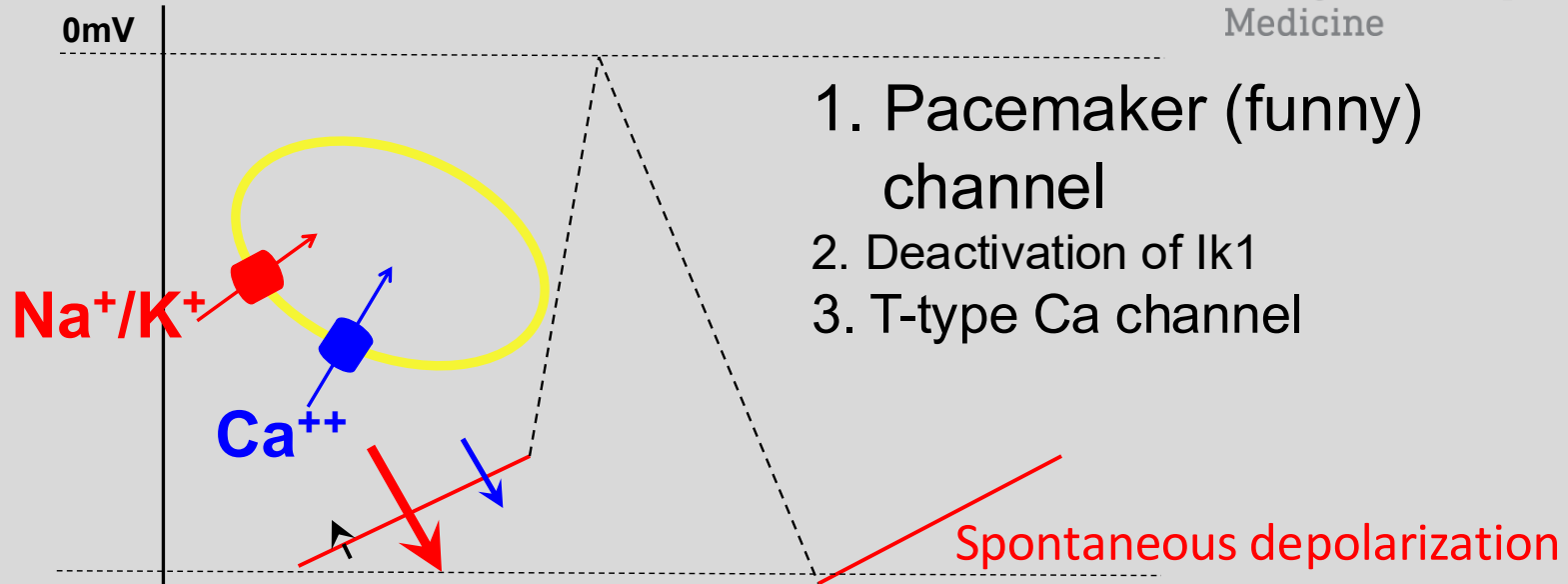
Resting Membrane Potential

Phase 4

No true resting membrane potential

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1. Pacemaker (funny) channel

2. Deactivation of I_{K1}

3. T-type Ca channel

If = hyperpolarization-activated non-selective cation current or “funny” current (Pacemaker current)

I_{K1} = deactivation of rectifying current;

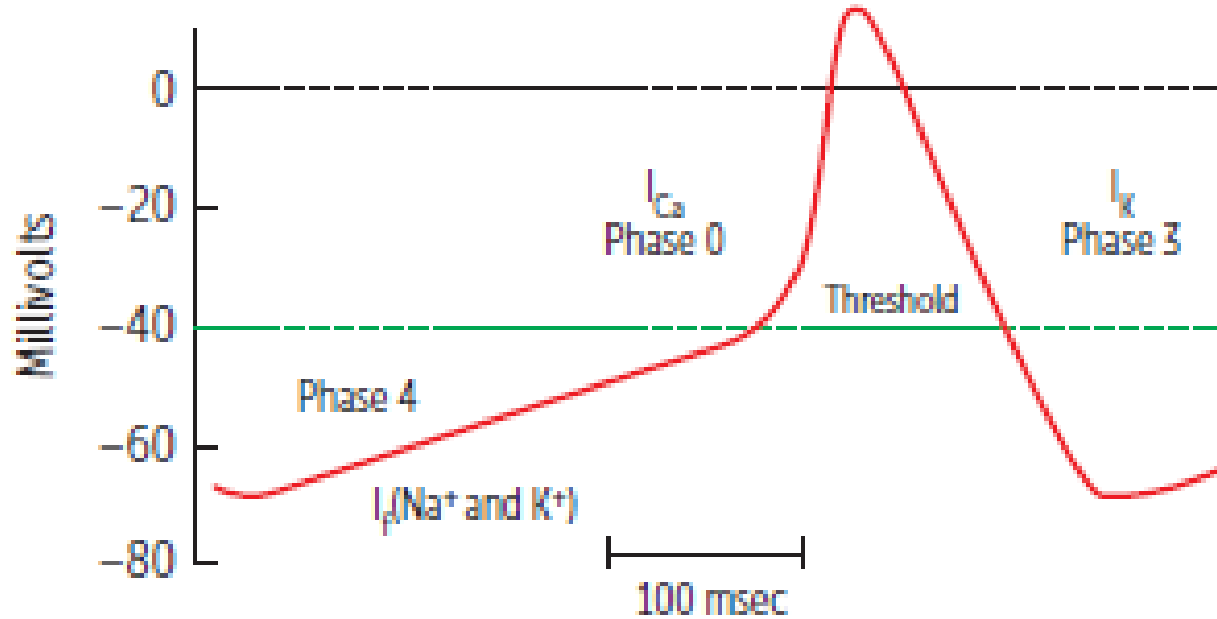
I_{Ca-T} = T-type calcium channel current.

If-mixed Na⁺/K⁺ inward current

Minimum requirement

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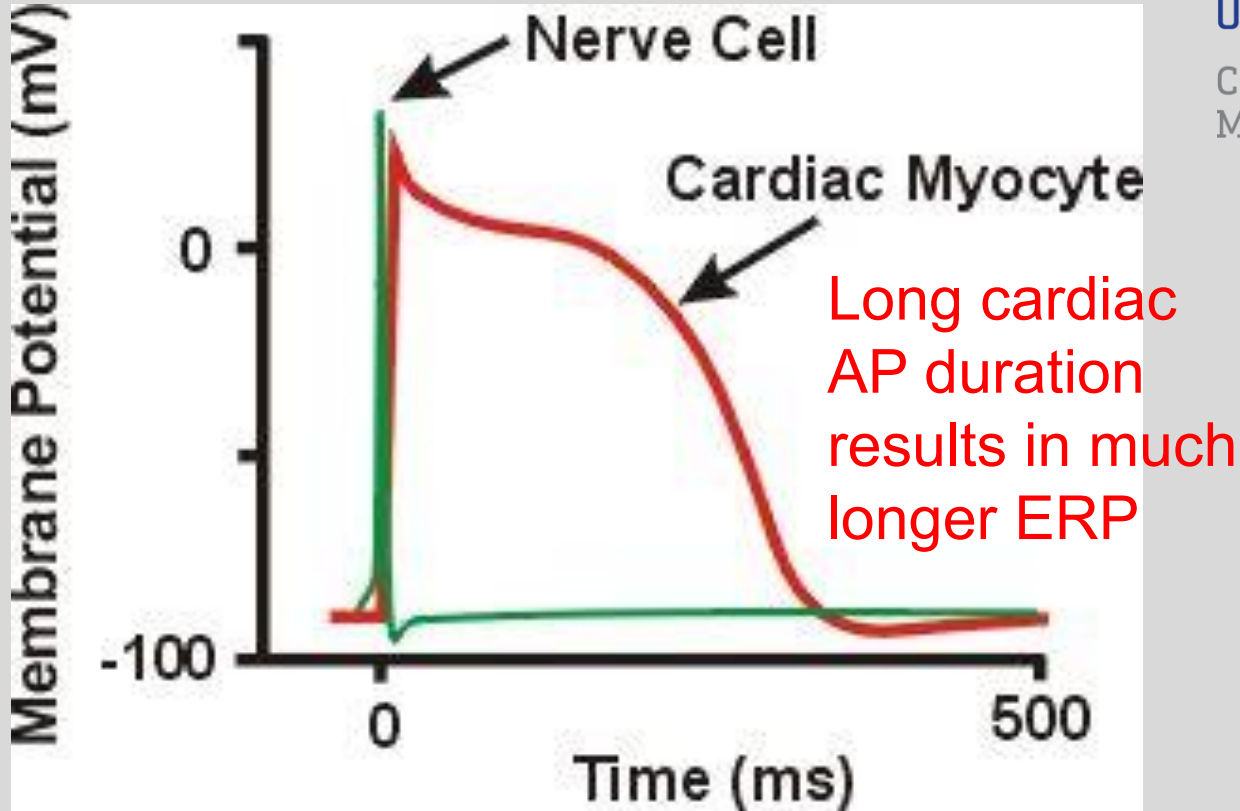
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Cardiac vs Nerve/Skeletal Muscle AP

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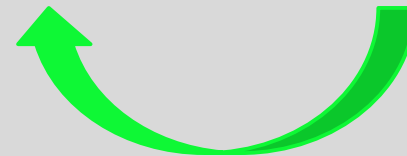
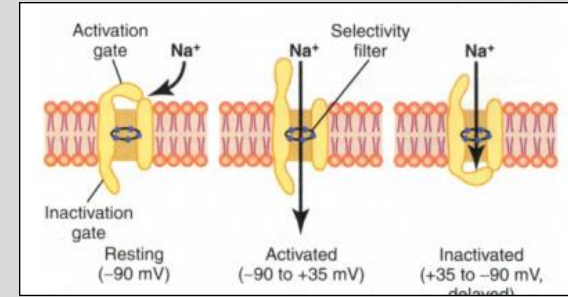
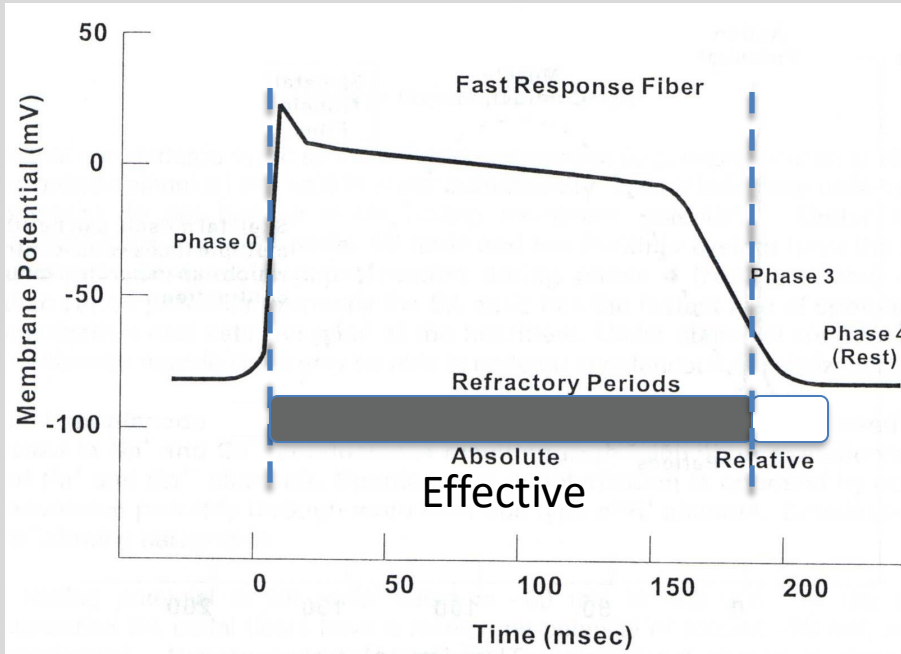
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Effective Refractory Period (ERP)

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1. **Effective refractory period (ERP)**
(Absolute refractory period)
No AP can be generated
Na⁺ channels: inactivated

2. **Relative refractory period (RRP)**
Stronger than normal stimulation
can generate a second, but smaller AP
Na⁺ channels: partially recovering

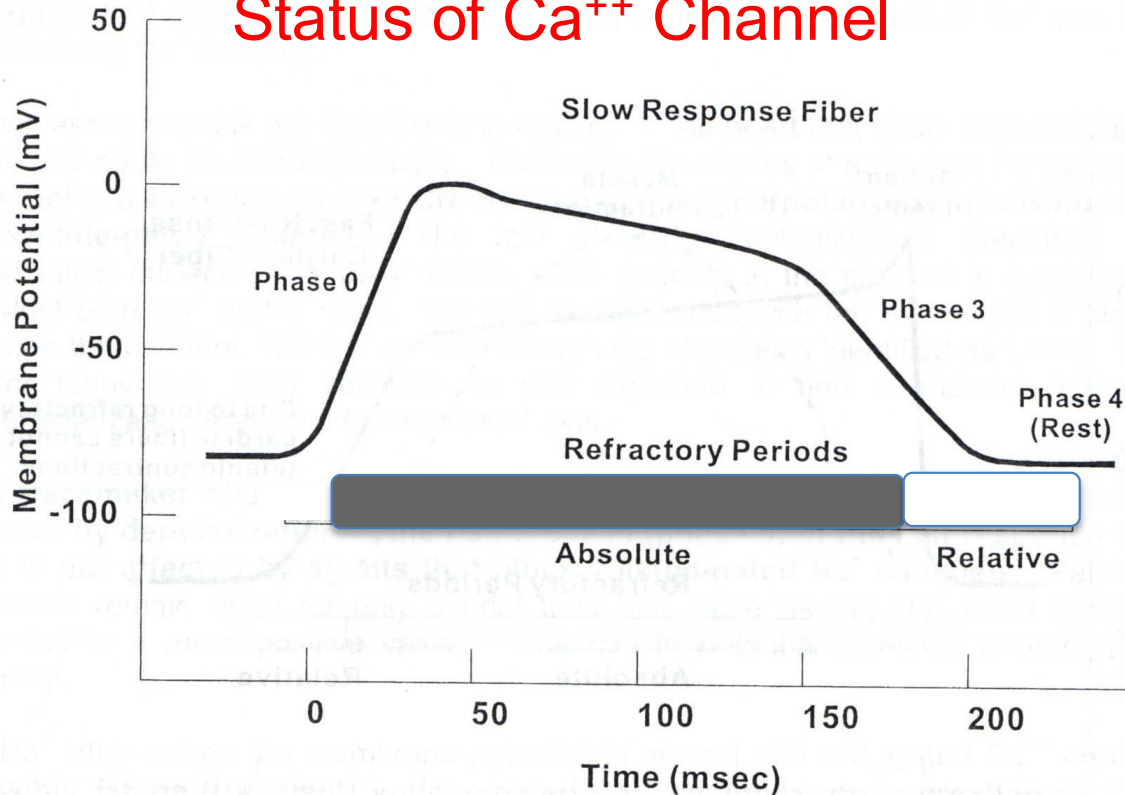
Effective Refractory Period (ERP)

Slow response fibers

Status of Ca^{++} Channel

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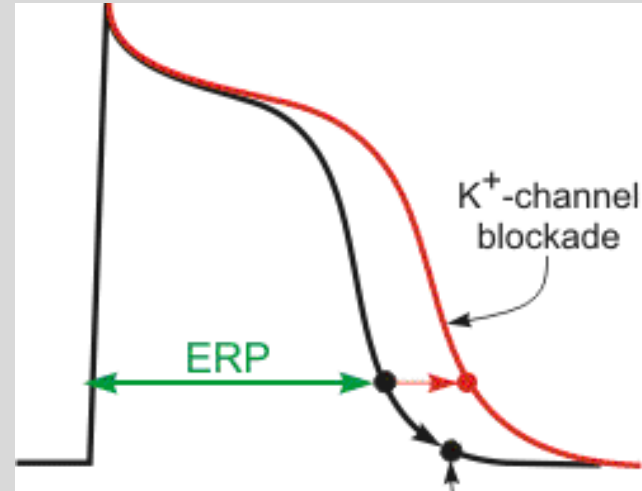
Effective Refractory Period (ERP)

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- ERP (ARP in other places) means the cell is refractory to the initiation of new, propagated action potentials.
- ERP is an important concept in arrhythmia formation and termination

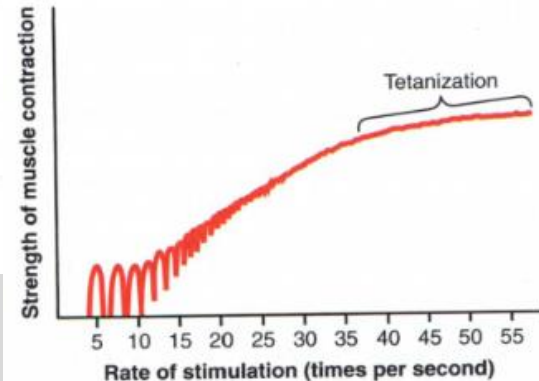
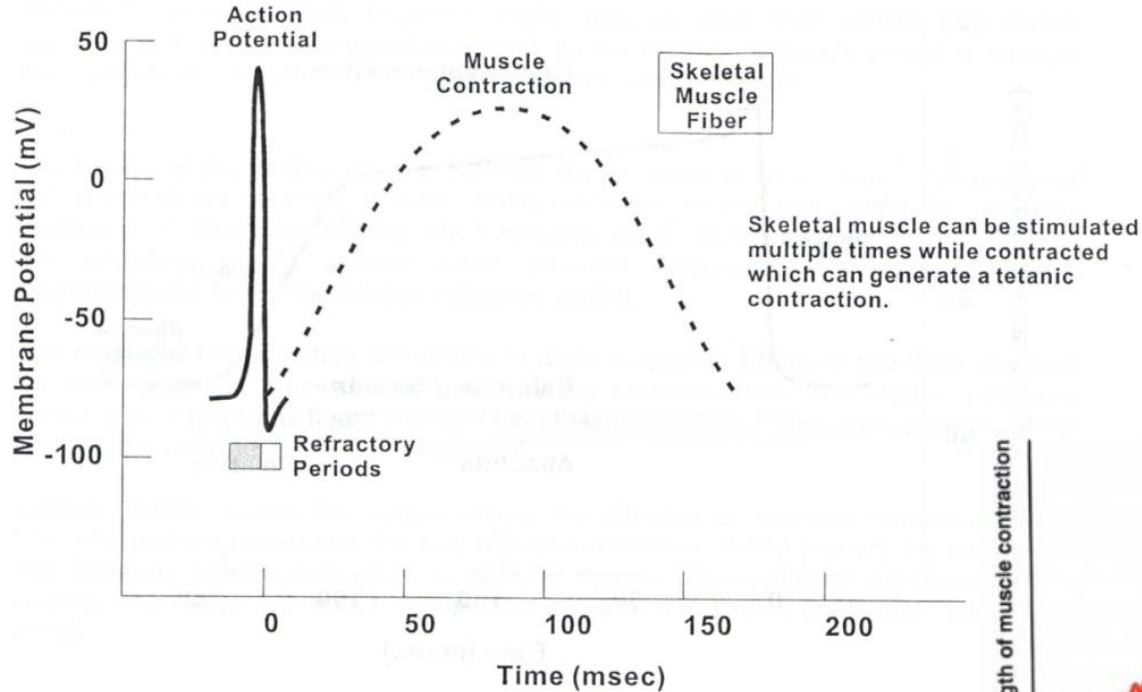
Generally ERP is correlated
to the AP duration



Tetanzation of skeletal muscle due to short ERP

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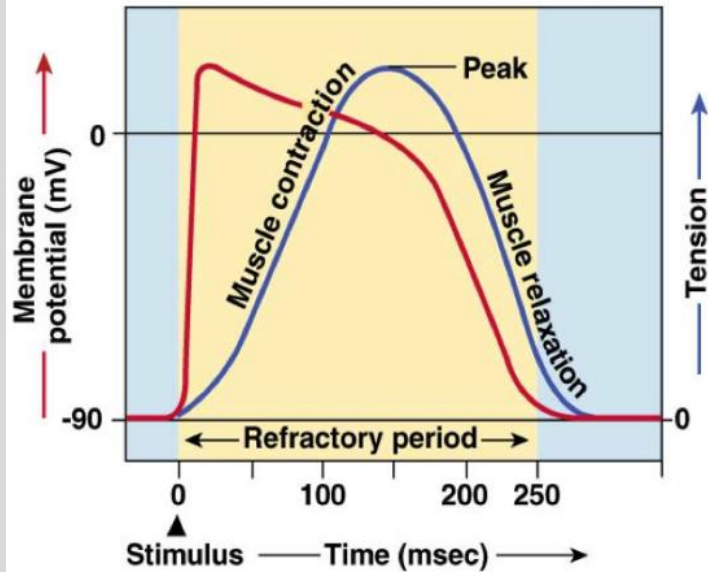
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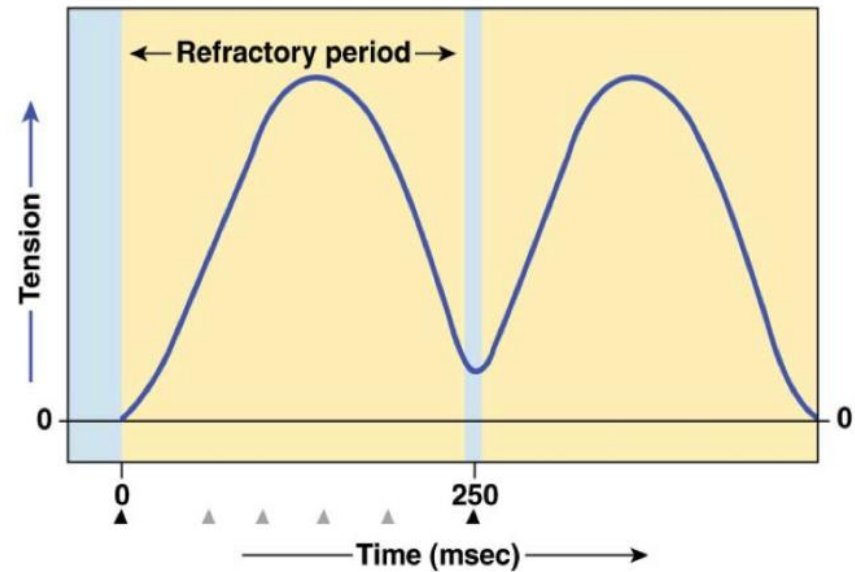
Long cardiac ERP prevents tetanus

Cardiac Muscle

(a) Cardiac muscle fiber: The refractory period lasts almost as long as the entire muscle twitch.



(b) Long refractory period in a cardiac muscle prevents tetanus.



Summary Slide

- 2 types of cardiac AP, 5 phases vs 3 phases for the fast vs slow AP
- Ion channels and currents in each phases of AP
- E-C coupling
- ERP
- Effect of K^+ concentration on cardiac AP
- **Core References:** Guyton and Hall Textbook of Medical Physiology, 14e, Chapter 9-from Section titled: Action Potentials in Cardiac Muscle to Section titled: Duration of Contraction

Some review questions

- What is/are the cause(s) of phase 2 (plateau phase) in fast response AP?
- Which channel blocker can slow depolarization in ventricular myocytes?
- Which channel blocker can slow depolarization in the AV nodal fibers?
- Low K^+ concentration can increase or decrease the duration of ventricular AP, why?

Lecture and Lab Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>